

Constraining the Origins of High-Redshift Supermassive Black Holes: A Comparative Study of Primordial and Stellar Seed Growth in JWST Little Red Dots

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Abstract

Recent observations from the James Webb Space Telescope (JWST) illustrate compact, high redshift “Little Red Dots” (LRDs)[6]. These LRDs reveal unexpectedly massive black holes less than one billion years after the Big Bang [14, 17, 18]. This rapid assembly challenges conventional models of black hole growth from stellar remnant seeds, which require continuous Eddington, and even super-Eddington accretion over cosmic timescales. Here we investigate whether primordial black holes (PBHs) that formed from density fluctuations in the early universe could serve as heavier initial seeds for these systems, enabling them to reach supermassive scales more rapidly without Eddington accretion rates [5]. Using JWST photometric and spectroscopic catalogs of LRDs and adopting analytic Eddington limited and super-Eddington growth models, our work compares the required seed masses, accretion rates, and general cycles under stellar seed and PBH-seed scenarios. Predicted luminosities and Eddington ratios are compared with multi-wavelength constraints from Chandra X-ray and ALMA radio surveys to test the plausibility of each growth pathway.

1 Introduction

The discovery of compact, infrared-bright sources known as *Little Red Dots* (LRDs) in JWST observations has amplified a long-standing question in astrophysics: how did supermassive black holes (SMBHs) form so quickly in the early universe? [4, 6, 14, 17, 18] These objects, found at redshifts $z > 6$, appear to host black holes with masses of $10^7 - 10^9 M_\odot$, existing less than one billion years after the Big Bang [6, 18]. Their inferred luminosities and number densities challenge the standard picture of black hole growth from stellar-remnant seeds—collapsed Population III stars with typical masses of $\sim 10^2 M_\odot$. Such seeds require nearly continuous Eddington or super-Eddington accretion to reach observed SMBH scales within the limited cosmic time available. Preliminary modeling suggests that stellar-seed growth is unlikely to be the starting point for the most luminous LRDs that we can observe unless we apply super-Eddington accretion constraints [8].

An alternative explanation posits that a substantial fraction of early SMBHs may have originated from primordial black holes (PBHs), hypothetical relics of high-density fluctuations in the early universe (within seconds after inflation) [3, 20, 23]. PBHs could serve as sufficiently massive initial seeds ($10^4 - 10^6 M_\odot$), reducing the need for prolonged and extreme accretion rates. Moreover, PBH seeding could help explain the unexpectedly high abundance of early AGN-like sources observed by JWST. Distinguishing between these pathways carries significant implications for cosmology, linking the microphysics of the early universe to the large-scale structures observed today [14, 18].

In this work we aim to compare the growth feasibility of stellar-remnant and primordial black hole seeds by applying analytic accretion models to JWST LRD data. By combining Eddington and super-Eddington growth scenarios with multi-wavelength constraints from *Chandra* and *ALMA*, this study evaluates which seeding mechanism more effectively reproduces the observed properties of high-redshift LRDs. The results are expected to provide new insight into the origin of the first SMBHs of the universe and to identify observational signatures that can further test these models with future data sets.

1.1 Little Red Dots

Little Red Dots (LRDs) were first found by JWST in 2022 and later confirmed in 2024. They are thought to be very compact galaxies at high redshifts. There are also theories that they are active galactic nuclei (AGN) because of their broad balmer emission lines [14, 17]. The reason they are so notable is because of the fact

that they house extremely massive black holes that appear so early in the universe. Little red dots also have a spike in abundance around 600 million years after the big bang but there are significantly less about 1.5 billion years after the big bang. This rapid increase and then disappearance is what makes little red dots so mysterious [17]. These extremely massive black holes would require stellar remnant seeds to grow at continuous Eddington and super Eddington rates for large timescales. This growth rate would likely lead to instability in the black hole’s growth. Primordial black hole seeds could also be used to model the growth of the black holes that could be present in LRDs. These seeds would not require as much Eddington accretion and could serve as a better explanation than stellar remnant seeds [5, 24].

In current literature, there is still disagreement on what physical component dominates the observed emission from LRDs. Some studies interpret them as AGN-like systems embedded in unusually dense gas that can both redden the observed spectrum and alter the emergent line profiles [19, 6]. Others emphasize that at least some LRDs show complex ultraviolet morphology and extended structure that can be consistent with compact galaxies undergoing intense star formation, potentially with an accreting black hole contributing part of the luminosity [17]. This ambiguity matters for our project because the inferred black hole mass and bolometric luminosity depend on modeling assumptions (e.g., whether broad lines trace virial motions around a black hole, whether the continuum is AGN-dominated, and how much dust or dense gas modifies the spectrum). As a result, our comparative growth analysis is best interpreted as a feasibility test: given a reasonable range of published M_{BH} and L_{bol} estimates for LRDs, we assess which seed scenarios require extreme or fine-tuned accretion histories and which scenarios more naturally match the implied growth constraints [5, 24].

Another reason LRDs are important is that surveys are beginning to show they are not isolated objects. Photometric selections in JWST blank fields have identified substantial candidate samples across $4 < z < 9$, suggesting a potentially significant number density of compact, red sources in the early universe [14]. If a nontrivial fraction of these objects host actively accreting black holes, then the existence of many luminous systems at early times tightens the “time budget” problem: stellar-remnant seeds would have to grow efficiently in many halos, not just in rare special cases. This motivates exploring heavier seeding channels, including primordial black holes, as a way to reduce the required growth factor between seed formation and the epoch when LRDs are observed.

1.2 Standard Inflation

Standard inflation is the leading theoretical framework for describing the earliest moments of the universe and provides the foundation for understanding the origin of primordial density fluctuations. It was originally proposed to resolve several conceptual problems of the classical Big Bang model, including the horizon problem, flatness problem, and the origin of large-scale structure [2, 21]. In addition to addressing these issues, inflation offers a causal mechanism for generating the small primordial perturbations that later grow into galaxies, galaxy clusters, and, under certain conditions, primordial black holes.

Inflation posits that the universe underwent a brief period of accelerated expansion at very early times, typically driven by the potential energy of a scalar field known as the inflaton. During this phase, the scale factor grew approximately exponentially,

$$a(t) \propto e^{Ht},$$

where H is the Hubble parameter, which is nearly constant during inflation [21]. This rapid expansion causes physical length scales to grow much faster than the Hubble radius H^{-1} , dramatically altering the causal structure of spacetime. Regions that were initially microscopic are stretched to astrophysical and even cosmological sizes.

One of the most important consequences of inflation is its effect on quantum fluctuations. In quantum field theory, fields cannot remain perfectly uniform; instead, they exhibit unavoidable vacuum fluctuations. During inflation, these fluctuations are continuously generated on sub-horizon scales, with typical amplitudes set by the Hubble scale during inflation. As the universe expands, these fluctuations are stretched beyond the Hubble radius, at which point causal microphysical processes can no longer affect them [2]. Once outside the horizon, their amplitudes effectively freeze, preserving information about the inflationary dynamics.

These frozen fluctuations correspond to perturbations in both the inflaton field and the spacetime metric, which together produce curvature perturbations. A key result of inflationary cosmology is that these curvature perturbations remain conserved on super-horizon scales for adiabatic perturbations. This conservation allows inflationary predictions to be carried forward reliably from the end of inflation through later cosmic epochs, including reheating, radiation domination, and matter domination [2].

After inflation ends, the universe transitions to a hot, radiation-dominated phase through a process known as reheating. During this period, the energy stored in the inflaton field is converted into ordinary particles, restoring the conditions required for standard Big Bang evolution. As the expansion rate slows, the Hubble radius

begins to grow faster than physical length scales. As a result, previously frozen perturbation modes eventually re-enter the horizon, at which point they can evolve dynamically under gravity and pressure forces [21].

The statistical properties of these primordial perturbations are commonly characterized by the power spectrum of curvature perturbations. In standard single-field slow-roll inflation, this power spectrum is nearly scale invariant, meaning that fluctuations have approximately the same amplitude on all length scales. This prediction is in excellent agreement with observations of the cosmic microwave background [20]. However, the near scale invariance of the spectrum also implies that typical density perturbations are very small, with amplitudes of order 10^{-5} , far below the threshold required for primordial black hole formation [3].

This limitation is crucial for the present project. While standard inflation successfully explains the observed large-scale structure of the universe, it does not naturally produce the large, localized density fluctuations needed to form primordial black holes without additional mechanisms. As a result, PBH formation requires either departures from simple slow-roll inflation or post-inflationary processes that selectively enhance fluctuations on specific scales [3, 20]. Understanding standard inflation therefore provides the baseline against which such enhancements can be identified and evaluated.

In summary, standard inflation establishes the causal origin and evolution of primordial perturbations, explains their near scale-invariant spectrum, and determines the background against which PBH formation scenarios must operate. Any viable mechanism for producing primordial black holes massive enough to seed early supermassive black holes must be consistent with the successful predictions of inflation on large scales while modifying its behavior on smaller, PBH-relevant scales [20].

1.3 Double Inflation Model

This section will explain the double inflation model which describes how there were two inflationary periods during the early universe. The first inflation generated the large-scale Cosmic Microwave Background fluctuations [3]. Followed by the preheating stage where the energy from the inflation field is converted into particles like quarks and leptons through parametric resonance [7]. This leads into a radiation dominated era and first inflation starts oscillating. Parametric resonance amplifies fluctuations at a specific k-mode. Parametric resonance is known as the periodic changes in an early universe field that amplify perturbations or produce particles exponentially, crucial for inflation, reheating, and primordial black hole formation, often described by the Mathieu equation, leading to rich phenomena like resonant

particle production, gravitational waves, and primordial non-Gaussianities seen in the cosmic microwave background. The second inflation exponentially stretches the amplified modes to cosmological scales. Re-enter the horizon later and potentially collapse into PBHs. [24]

A crucial advantage of this picture for this work is that it can generate a relatively narrow preferred PBH mass range. Because parametric resonance amplifies only a band of modes, it effectively selects a characteristic comoving scale k_* . That scale later corresponds to a particular horizon mass at re-entry and therefore to a characteristic PBH mass. This is relevant to Little Red Dots because the seeding scenario is most effective if PBHs are not predominantly tiny (which would not help with SMBH seeding) nor predominantly too massive (which could violate abundance constraints). Instead, a peak near 10^4 – $10^6 M_\odot$ is attractive because it reduces the required number of e-foldings of growth to reach the SMBH masses inferred for LRDs [24, 5]. In the literature, this motivates targeting models that can produce a sharply peaked small-scale power spectrum while leaving the CMB-scale spectrum nearly unchanged, since the latter is tightly constrained by observations.

1.4 Primordial Black Hole Formation

Primordial Black Holes (PBHs) are theorized to have been formed by the collapse of large over-dense regions under gravity shortly after entering the horizon during the radiation-dominated era [3, 20, 23]. A Friedmann–Lemaître–Robertson–Walker model (FLRW) is used to define early-universe dynamics that foster the proper conditions for PBH formation. The FLRW metric describes a homogeneous, isotropic, expanding (or otherwise, contracting) universe that is path-connected, but not necessarily simply connected. Let us write the line element of an FLRW model as follows:

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - Kr^2} + r^2 d\Omega^2 \right], \quad (1)$$

where $a(t)$, K and dW are the scale factor, spatial curvature, and line element of the unit two-sphere, respectively. To define spatial curvature, K is a parameter that describes the intrinsic geometry of spatial slices in the universe. The three possible values for this parameter, 1, 0, or -1, define the spatial geometry of the universe as spherical, flat, or hyperbolic. The relevant Friedmann equation is given by:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3}\rho - \frac{K}{a^2}, \quad (2)$$

where the dot “.” denotes the time derivative. Introducing the areal radius $R := a(t)r$, the above equation can be rewritten as follows:

$$\frac{\dot{R}^2}{2} - \frac{M}{R} = E(r) := -\frac{Kr^2}{2}, \quad (3)$$

where M is the mass inside the co-moving radius r given by $\frac{4\pi\rho R^3}{3}$. This equation can be regarded as the energy conservation law for a fluid element on the co-moving radius r , relevant for defining the conditions of the universe in which PBHs will form in. We now assume that the background universe is given by a spatially flat ($K = 0$) FLRW universe. In this context, we now suppose the perfect fluid, a theoretical fluid model characterized with zero viscosity, simplifying fluid dynamics, with a linear equation of state is given by $p = w\rho$ with a constant w , where ρ and p are the energy density and the pressure. The constant w is given by $\frac{1}{3}$, specifically for the radiation-dominated (RD) universe, the time period when PBHs are theorized to have been formed in due to the high density and pressure in the universe at the time. From the conservation law of the fluid $d(\rho a^3) = \rho da^3$, we obtain the following.

$$\rho \propto a^{-3(1+w)}, \quad (4)$$

Then, we find the behavior of the Hubble radius as:

$$R_H := \frac{1}{H} \propto a^{3(1+w)/2} \quad (5)$$

This model and equation, therefore, allow for the existence of density perturbations. This law specifically allows for density perturbations because it is a local conservation equation that holds for each fluid element independently, but the universe need not be perfectly homogeneous. In a strictly homogeneous background, density evolves uniformly everywhere. However, small initial inhomogeneities (e.g., from quantum fluctuations during inflation) introduce spatial variations. During this early period, all relevant perturbations have super-horizon scales. The amplitude of the curvature perturbation is conserved on super-horizon scales, so the fluctuations generated in the inflationary era are “frozen” after horizon exit and start to grow after horizon re-entry [3, 23].

Because we are only considering formation during the RD universe, perturbations on super-horizon scales are considered for initial fluctuations of the subsequent evolution of an over-dense region. Let us take uniform Hubble time slices and compare the Friedmann equation in the over-dense region and the flat background. On a

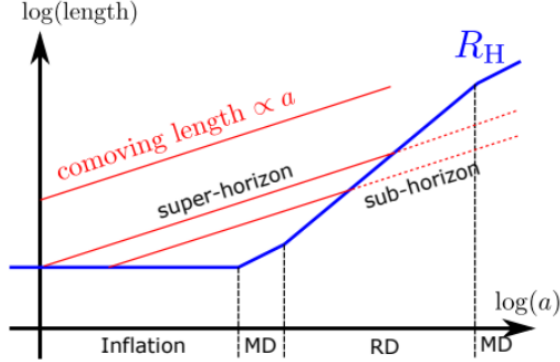


Figure 1: The rough sketch of the standard evolution of the background universe and the perturbations on it.

uniform Hubble time slice, since the value of the Hubble expansion rate is identical, we obtain:

$$\frac{8}{3}\pi\rho_b = \frac{8}{3}\pi\rho - \frac{K}{a^2}, \quad (6)$$

where ρ and ρ_b denote the density in the over-dense and background (surrounding universe) regions, respectively. The over-dense region is described by the closed FLRW with the spatial curvature K . This density perturbation can therefore be calculated as:

$$\delta = \frac{\rho - \rho_b}{\rho_b}. \quad (7)$$

where δ is a scalar value representing the amplitude of the density perturbation. Let $\bar{\delta}$ be the amplitude of the volume average of the density perturbation at horizon re-entry. If $\bar{\delta}$ exceeds a certain threshold value, PBH formation is realized [23].

The primary reason this formalism matters for our study is that it makes explicit the bridge between early-universe fluctuation physics and the seed masses required by the LRD problem. In PBH scenarios, the mass of a PBH is not an adjustable astrophysical parameter in the same way that a stellar remnant mass might be; instead, it is tied to the horizon scale at the time the responsible perturbation re-enters and collapses. This means that any model aiming to produce PBH seeds in the 10^4 – $10^6 M_\odot$ range must enhance perturbations on a specific range of comoving scales and must do so in a way that is consistent with observational constraints on the power spectrum at other scales [23, 3, 20]. In other words, PBH seeding is predictive: once a formation mechanism is specified, it constrains both the typical PBH masses

and their expected abundance.

This perspective also clarifies why PBHs are often discussed as “rare peaks” rather than typical outcomes. The threshold condition $\bar{\delta} > \delta_c$ implies that only a small fraction of regions collapse, so PBHs form from the extreme tail of the fluctuation distribution. As a result, PBH abundances can change dramatically depending on the variance and any non-Gaussian features of the primordial fluctuations [23]. For our project, this is conceptually important because it sets expectations for whether PBHs could plausibly seed a significant fraction of LRDs: even if PBHs form, they may be too rare to explain the observed number density unless the enhancement mechanism produces a sufficiently high formation probability. Our modeling approach therefore treats PBHs as a candidate seeding channel whose viability depends on both growth feasibility and population-level plausibility when compared to LRD demographics [5, 14, 24].

2 Technical Details

This study employs analytic and semi-empirical modeling to compare black hole growth from stellar-remnant seeds and primordial black hole (PBH) seeds in the context of JWST “Little Red Dots” (LRDs). The analysis combines publicly available photometric and spectroscopic data from JWST deep-field surveys with basic accretion physics to evaluate the feasibility of each seeding scenario within cosmic time constraints.

For each LRD, the starting parameters include the observed redshift, bolometric luminosity, and estimated black hole mass derived from published emission-line widths or luminosity scaling relations. The Eddington Luminosity is the maximum luminosity a celestial body can achieve and is defined as

$$L_{\text{Edd}} = 1.26 \times 10^{38} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) \text{ erg s}^{-1}, \quad (8)$$

The Eddington luminosity plays a central role in our project because it translates an observed black hole mass into a physically motivated upper bound on the steady luminosity produced by accretion. The physical meaning of L_{Edd} is that it arises from balancing two competing effects on ionized gas: gravity pulls matter inward toward the black hole, while radiation emitted by the accretion flow pushes outward on free electrons (and thus on the gas) through Thomson scattering. In the expression, M_{BH} is the black hole mass and $M_{\odot}$ is the solar mass used as a standard unit. The

constant coefficient collects fundamental constants including G , c , the proton mass, and the Thomson cross section. Because $L_{\text{Edd}} \propto M_{\text{BH}}$, more massive black holes can shine more brightly without blowing away their fuel supply. In our comparison of seed scenarios, this scaling matters because heavier initial seeds can remain at moderate Eddington ratios while still producing large luminosities, whereas lighter stellar seeds must spend longer periods accreting near or above the Eddington limit to reach the same observed masses.

The Eddington ratio λ_{Edd} is a dimensionless measure of how “aggressively” a black hole is accreting compared to the Eddington-limited case. In our work,

$$\lambda_{\text{Edd}} = \frac{L_{\text{bol}}}{L_{\text{Edd}}},$$

where L_{bol} is the bolometric luminosity inferred from JWST data (often requiring bolometric corrections and assumptions about whether emission is dominated by stellar light or AGN emission). Values of $\lambda_{\text{Edd}} \approx 1$ correspond to the canonical Eddington-limited thin-disk regime, while $\lambda_{\text{Edd}} > 1$ is interpreted as super-Eddington accretion or as evidence that the luminosity, mass, or bolometric correction has been mis-estimated. This parameter is directly useful for evaluating physical plausibility: if reproducing an LRD requires $\lambda_{\text{Edd}} \gg 1$ sustained over long times, that scenario may be difficult to realize for a large population of objects. Conversely, if a heavier seed mass brings the required λ_{Edd} into the range ~ 0.1 – 1 , the growth pathway becomes more natural [5, 13].

The Salpeter timescale t_S is important because it gives an intuitive time unit for exponential black hole growth. Under the assumption of radiatively efficient accretion, a fraction ϵ of the rest-mass energy of infalling gas is converted into radiation, and the remaining fraction $(1 - \epsilon)$ contributes to increasing the black hole mass. In the expression

$$t_S \approx 4.5 \times 10^7 \left(\frac{\epsilon}{0.1} \right) \text{ yr},$$

ϵ is the radiative efficiency, typically around 0.1 for a standard thin disk. The key interpretation is that at Eddington-limited accretion, the black hole mass increases by a factor of $e \approx 2.718$ every $\sim t_S$ years. Therefore, the total growth factor achievable between two cosmic times depends on how many Salpeter times fit into that interval. This is exactly what makes high-redshift black holes challenging: the universe is only a few hundred million years old at $z \geq 10$, which corresponds to only a handful of Salpeter times. This makes it difficult for $\sim 10^2 M_\odot$ stellar seeds to reach $10^7 M_\odot$ unless accretion is near-continuous and often super-Eddington, while heavier PBH seeds require fewer e-foldings to reach the same final masses [5, 24].

The exponential growth model

$$M(t) = M_{\text{seed}} \exp \left[\frac{(1 - \epsilon) \lambda_{\text{Edd}} t}{\epsilon t_S} \right]$$

is the main quantitative tool we use to compare seed scenarios. In this equation, $M(t)$ is the black hole mass after a time t has passed, M_{seed} is the initial mass of the seed black hole, λ_{Edd} is the assumed Eddington ratio during the growth period, ϵ is the radiative efficiency, and t_S is the Salpeter timescale. The physical content is that the growth rate is proportional to the current mass, so growth compounds: a black hole that is already more massive can grow more rapidly in absolute terms while maintaining the same Eddington ratio. This is why the initial seed mass is so impactful. For our project, the equation allows us to invert the problem for each LRD: given an observed M_{BH} at a known redshift, we can estimate the minimum λ_{Edd} required for a chosen seed mass, or alternatively the minimum seed mass required if λ_{Edd} is bounded by physical plausibility. This is the basis for deciding whether stellar seeds require extreme assumptions compared to PBH seeds [8, 13, 24].

For each LRD, the minimum combination of seed mass and accretion rate capable of reproducing the observed M_{BH} at its redshift is determined under both Eddington-limited ($\lambda_{\text{Edd}} = 1$) and super-Eddington ($\lambda_{\text{Edd}} > 1$) regimes. The predicted luminosities and timescales are then compared against multi-wavelength constraints derived using *Chandra* X-ray and *ALMA* radio archives to test for consistency with observed emission or non-detections.

Multi-wavelength constraints play an important role in interpreting the accretion states of LRDs. In particular, X-ray observations from *Chandra* and submillimeter observations from *ALMA* provide independent probes of black hole activity and obscured star formation. If an LRD hosts a black hole accreting at or above the Eddington limit, standard AGN models predict detectable X-ray emission unless the source is heavily obscured or the accretion flow is radiatively inefficient. Similarly, strong star formation or dust-reprocessed AGN emission is expected to produce submillimeter signatures detectable by ALMA. The absence of strong detections in some LRDs therefore places constraints on how extreme their accretion rates and star formation rates can be [6, 19].

For our project, these non-detections are not interpreted as definitive evidence against accretion, but rather as consistency checks. Growth scenarios that require long-lived, highly super-Eddington accretion for many objects may overpredict X-ray or submillimeter emission for a substantial fraction of the LRD population. In contrast, scenarios involving heavier initial seeds can reproduce the same final black hole masses with more moderate Eddington ratios, potentially remaining consistent

with current multi-wavelength limits. Thus, the inclusion of Chandra and ALMA constraints helps distinguish between growth pathways that are merely mathematically possible and those that are physically plausible given existing observational data [5, 13].

A key ingredient in evaluating the feasibility of any growth scenario is the amount of cosmic time available between seed formation and observation. At high redshift, cosmic time is a steep function of redshift, and small changes in z correspond to large fractional changes in the age of the universe. For a standard flat Λ CDM cosmology, the cosmic age at redshift z is given by

$$t(z) = \int_z^\infty \frac{dz'}{(1+z')H(z')},$$

where $H(z)$ is the redshift-dependent Hubble parameter. While this expression can be evaluated numerically, it is sufficient for our purposes to note that the universe is only ~ 270 Myr old at $z \approx 14$, ~ 480 Myr old at $z \approx 10$, and ~ 950 Myr old at $z \approx 6$. These timescales correspond to only a few to a few tens of Salpeter times, even under optimistic assumptions. This strong time constraint is what makes early supermassive black holes difficult to explain using stellar-remnant seeds alone, and it motivates exploring heavier primordial seeds that require fewer e-foldings of growth to reach the masses inferred for LRDs [5, 24].

All calculations are performed using analytic methods and publicly available data, emphasizing physical reasoning rather than numerical simulation. This framework provides an interpretable comparison between primordial and stellar seeding pathways for the early supermassive black holes powering the LRDs observed by JWST thus far.

Taken together, these analytic growth calculations provide a transparent framework for comparing stellar-remnant and primordial black hole seeding scenarios in the context of JWST Little Red Dots. Rather than attempting to model every detail of galaxy formation or accretion physics, this approach isolates the core physical tension: the combination of limited cosmic time, high inferred black hole masses, and constraints on sustained super-Eddington accretion. By mapping observed LRD properties onto required seed masses and Eddington ratios, we identify which regions of parameter space are viable and which require extreme or fine-tuned assumptions. This sets the stage for interpreting the results in a broader cosmological context and for identifying observational signatures that can further test primordial versus stellar seeding scenarios as JWST and multi-wavelength surveys continue to expand.

2.1 General Procedure and Analysis Outline

The objective of this project is to assess the physical plausibility of different black hole seeding scenarios for LRDs. Our approach is analytic and comparative, combining observational constraints from JWST with standard black hole growth models to evaluate whether stellar-remnant seeds or PBH seeds more naturally reproduce the inferred properties of these objects within the available cosmic time.

The analysis begins by compiling observational parameters for LRDs from recent JWST studies, including redshift estimates, bolometric luminosities, and published black hole mass estimates. Where uncertainties or multiple interpretations exist, such as whether the emission is dominated by an active galactic nucleus or by dense stellar populations, we adopt a reasonable range of values consistent with the literature. These quantities define the target final black hole masses and luminosities that any viable growth scenario must reproduce.

Next, we define initial conditions corresponding to different seeding pathways. Stellar-remnant seed models assume initial black hole masses of order $10^2 M_\odot$, motivated by the collapse of Population III stars, while PBH seed models assume initial masses in the range 10^4 – $10^6 M_\odot$, motivated by early-universe density fluctuation scenarios. For both cases, we assume seed formation occurs sufficiently early that the dominant constraint is the time available for growth prior to the observed redshift of each LRD.

Black hole growth is then modeled using analytic prescriptions based on Eddington-limited and super-Eddington accretion. Using the exponential growth framework, we compute the mass evolution of a black hole as a function of seed mass, Eddington ratio, radiative efficiency, and elapsed cosmic time. For each LRD, we either determine the minimum Eddington ratio required to grow from a given seed mass to the observed black hole mass by the observed redshift, or alternatively determine the minimum seed mass required if the Eddington ratio is constrained to remain within physically plausible bounds. This step directly quantifies how extreme the required accretion history must be under each seeding scenario.

To incorporate cosmological constraints, we compute the cosmic time available for growth using standard relations between redshift and the age of the universe in flat Λ CDM cosmology. Expressing growth in units of Salpeter times provides an intuitive measure of whether a given scenario requires near-continuous Eddington or super-Eddington accretion over an extended period or whether more moderate accretion histories are sufficient.

The inferred accretion states are then evaluated for consistency with existing multi-wavelength observations. X-ray constraints from *Chandra* and submillimeter constraints from *ALMA* are used as consistency checks, because long-lived, highly

super-Eddington accretion is expected to produce detectable high-energy or dust-reprocessed emission unless the source is extremely obscured. Growth scenarios that systematically require such extreme accretion for a large fraction of the LRD population are therefore treated with caution.

Finally, we compare the stellar-remnant and PBH seed scenarios across the full sample of LRDs, focusing on the required accretion rates, sensitivity to fine-tuning, and compatibility with observational constraints. Rather than attempting to determine a unique origin for each object, this analysis aims to identify which seeding pathway more naturally satisfies the combined constraints imposed by cosmic time, observed black hole masses, and current multi-wavelength data. This comparative framework allows us to assess the physical plausibility of PBHs as seeds for the early supermassive black holes inferred in JWST little red dots.

3 Discussion and Results

This work used the semi-analytical model A-SLOTH [10] [15]. It is used to simulate early stars and galaxies. Figure 2 shows the required Eddington Ratios for

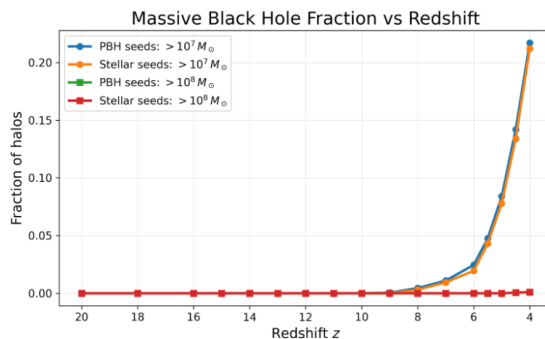


Figure 2: A graph of the fraction of halos

seeds of different masses, the stellar and primordial seeds. The stellar seeds require an Eddington ratio much closer to the Eddington limit

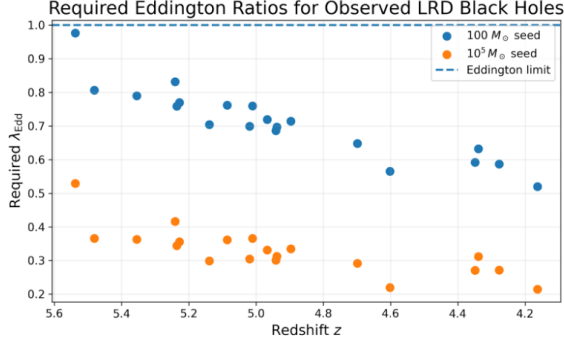


Figure 3: A graph of required Eddington ratios for black hole seed masses

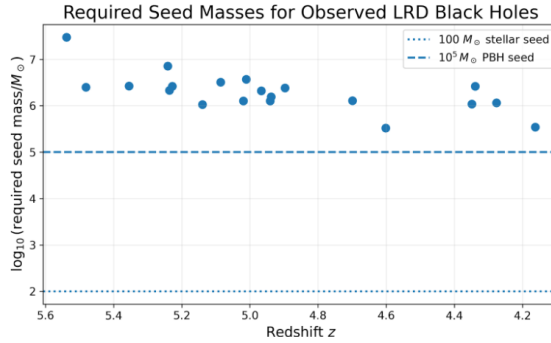


Figure 4: A graph of required seed masses for LRD black holes

4 Discussion and Results

To investigate whether stellar-remnant or primordial black hole seeds more naturally reproduce the observed properties of JWST Little Red Dots (LRDs), we modeled black hole growth histories using analytic accretion prescriptions and semi-analytic structure formation methods inspired by the A-SLOTH framework [10, 15, 12]. Our analysis focused on comparing the growth requirements imposed by observed LRD black hole masses under different assumptions regarding seed mass, Eddington ratio, duty cycle, and available cosmic time.

The rapid appearance of supermassive black holes in the early universe has become one of the major unresolved problems in modern astrophysics [22, 11]. Recent JWST observations have revealed a large population of compact high-redshift systems known as Little Red Dots, many of which are believed to host actively accreting supermassive black holes [9, 1]. These observations place strong constraints on viable

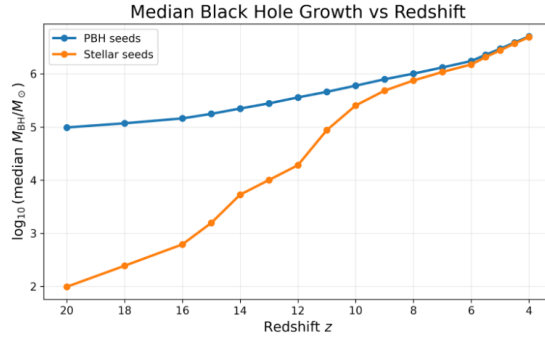


Figure 5: A graph of median black hole growth for both seeds

black hole growth histories because the universe was still less than one billion years old at the redshifts where many LRDs are observed.

The A-SLOTH framework models the evolution of early galaxies and black holes using extended Press-Schechter (EPS) merger trees, allowing dark matter halo growth histories and hierarchical mergers to be tracked efficiently over cosmic time [12]. Semi-analytic approaches provide an efficient method for exploring large cosmological parameter spaces while incorporating realistic halo growth histories, merger trees, accretion physics, and feedback regulation [12]. This approach is especially useful for investigating the physical plausibility of different seed formation scenarios while remaining computationally feasible. In these models, black hole growth occurs primarily through gas accretion, while mergers contribute secondarily to the final SMBH mass budget [12]. This distinction is important because it implies that the feasibility of early SMBH formation depends more strongly on sustained accretion conditions than on rare merger events alone.

This work used the semi-analytical model A-SLOTH [10, 15], which is designed to simulate the formation of the first stars, galaxies, and black holes in the early universe. The framework incorporates dark matter halo growth, hierarchical structure formation, gas accretion, halo mergers, and black hole feedback effects in order to evaluate realistic black hole growth histories. By combining these semi-analytic methods with analytic accretion calculations, we compared the ability of stellar-remnant and primordial black hole seeds to reproduce the observed masses of LRD black holes.

Figure 2 shows the required Eddington ratios necessary for black holes to grow from different initial seed masses into the observed LRD black holes by their measured redshifts. Stellar-remnant seeds with initial masses near $10^2 M_{\odot}$ generally re-

quire Eddington ratios very close to or exceeding unity for prolonged periods of time. Sustained near-Eddington accretion is difficult to maintain because radiation pressure, feedback effects, and finite gas supply can suppress long-term black hole growth [16, 11]. Sustained accretion at or above the Eddington limit is physically difficult because radiation pressure opposes inflowing gas and can suppress continued accretion. In addition, strong feedback from black hole radiation, jets, and winds may heat surrounding gas and reduce the available fuel supply [12].

By contrast, primordial black hole seeds with masses in the range 10^4 – $10^6 M_\odot$ require significantly lower Eddington ratios to reproduce the same observed systems. This reduction in the required accretion rate makes PBH seed scenarios substantially more plausible under realistic physical conditions. Heavy-seed formation channels, including primordial black holes and direct-collapse black holes, have therefore been proposed as possible explanations for rapid SMBH assembly at high redshift [22, 11]. The results therefore support the idea that larger initial seed masses alleviate the severe time constraints imposed by the early universe.

Figure 3 illustrates the minimum seed masses required to reproduce observed LRD black holes under physically reasonable accretion assumptions. Most inferred seed masses fall between approximately 10^5 and $10^7 M_\odot$. These masses are substantially larger than conventional Population III stellar remnants and instead align more naturally with heavy-seed formation channels such as primordial black holes or direct-collapse black holes (DCBHs). Similar conclusions regarding the difficulty of reproducing observed LRD populations with stellar-remnant seeds alone have also been found in recent semi-analytic SMBH population studies [12, 5].

Figure 4 compares the median black hole growth histories for stellar-remnant and primordial black hole seeds. The PBH growth tracks reach supermassive scales substantially earlier and with less extreme growth histories than the stellar-remnant models. Stellar seeds frequently fail to reproduce the highest observed LRD black hole masses within the available cosmic time unless continuous near-Eddington or super-Eddington accretion is assumed. This result highlights the core “cosmic timing problem” motivating this work.

The inclusion of halo merger trees and hierarchical structure growth further strengthens this conclusion. In realistic cosmological environments, halos undergo mergers, tidal interactions, and feedback-regulated star formation throughout their evolution. Black hole growth is therefore not perfectly smooth or uninterrupted. Jeon et al. show that black hole accretion is often limited by the available cold gas supply and duty cycle effects, meaning that even if high accretion rates are theoretically possible, they may not persist continuously over long timescales [12]. As a result, stellar-remnant seeds face even greater difficulty reaching observed LRD

masses under realistic astrophysical conditions.

Feedback processes also play an important role in regulating growth. As black holes accrete matter and emit radiation, surrounding gas can be heated and partially expelled from the halo environment. This feedback reduces subsequent accretion efficiency and limits long-term growth [12]. Heavy seed models are less affected by this limitation because they require fewer e-foldings of exponential growth to reproduce observed SMBH masses.

Our results are also broadly consistent with current multi-wavelength observational constraints. Extremely prolonged super-Eddington accretion would likely produce stronger X-ray and submillimeter emission than is observed for many LRD systems. PBH seeds, which require more moderate accretion histories, remain more compatible with current *Chandra* and *ALMA* limits.

However, several uncertainties remain. Black hole masses for LRDs are still difficult to estimate reliably because current methods rely on indirect spectral scaling relations calibrated primarily at lower redshifts [9]. Additionally, the true fraction of LRDs powered by AGN activity versus compact star formation remains uncertain. Our conclusions should therefore be interpreted as a comparative feasibility analysis rather than definitive proof of primordial black hole seeding.

Overall, the results consistently indicate that heavy-seed scenarios, particularly primordial black hole seeds, provide a more physically plausible pathway for producing the massive black holes inferred in JWST Little Red Dots within the limited cosmic time available after the Big Bang.

5 Conclusion

The discovery of JWST Little Red Dots has intensified the challenge of explaining how supermassive black holes formed so rapidly in the early universe. The inferred black hole masses and number densities of these systems place strong constraints on viable growth histories and seeding mechanisms.

In this work, we compared stellar-remnant and primordial black hole seed scenarios using analytic accretion models, cosmological growth constraints, and semi-analytic structure formation methods motivated by the A-SLOTH framework. Our analysis demonstrates that stellar-remnant seeds generally require prolonged near-Eddington or super-Eddington accretion in order to reproduce the observed LRD black hole population within the available cosmic time. Such sustained growth histories become increasingly difficult under realistic astrophysical conditions involving feedback, finite gas supply, and intermittent accretion.

Primordial black hole seeds, by contrast, naturally reduce the required growth factor by beginning with substantially larger initial masses. As a result, PBH seed models reproduce observed LRD black hole masses under significantly more moderate accretion conditions and remain more consistent with current multi-wavelength observational constraints.

These findings suggest that heavy-seed formation channels may play an important role in the origin of the earliest supermassive black holes. While current uncertainties in LRD black hole mass estimates and AGN classifications remain significant, future JWST spectroscopy, deep X-ray observations, and improved cosmological simulations will further constrain the viability of primordial black hole seeding scenarios.

Future JWST spectroscopy and multi-wavelength observations will be critical for distinguishing between competing early SMBH formation scenarios [1, 9]. Ultimately, the origin of early supermassive black holes remains one of the most important open questions in modern astrophysics. JWST observations now provide an unprecedented opportunity to connect early-universe cosmology, black hole physics, and galaxy formation through the study of Little Red Dots.

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A General Relativity and its Relevance to Primordial Black Holes

General Relativity arises from the need to understand how gravity operates in situations where classical Newtonian physics breaks down. In the early universe, matter densities were extraordinarily high, particles moved at relativistic speeds, and spacetime itself was rapidly evolving. Under these conditions, Newtonian gravity is no longer sufficient, and a relativistic description of gravity becomes necessary. Since primordial black holes (PBHs) are hypothesized to form during this early epoch, any physically consistent description of their formation must be grounded in relativistic physics.

A useful starting point for understanding relativity is electromagnetism. In classical physics, the motion of a charged particle in an electromagnetic field is governed by the Lorentz force law,

$$F = q(E + v \times B),$$

which describes how particles respond to electric and magnetic fields. However, the Lorentz force law alone does not describe how the fields themselves form or how they change. That information is contained in Maxwell's equations, such as Gauss's law and Ampère's law, which relate electric and magnetic fields to their sources. This distinction highlights an important conceptual point: force laws describe how particles move within a background field, but additional structure is required to describe the field itself.

A similar issue arises in gravity. Newton's law of gravitation describes how masses attract each other, but it assumes a fixed background of space and time. When gravity becomes strong, or when velocities approach the speed of light, this assumption fails. Special Relativity resolves inconsistencies between mechanics and electromagnetism by unifying space and time into a single four dimensional spacetime. However, Special Relativity does not explain gravity, because it still assumes spacetime itself is fixed and unaffected by matter.

General Relativity extends this idea by allowing spacetime to become dynamical. Instead of treating gravity as a force acting within spacetime, it is described as a manifestation of spacetime curvature. Matter and energy determine how spacetime curves and objects move along paths determined by that curvature. This geometric description replaces the concept of gravitational force. In this framework, the motion of particles is no longer governed by a force law such as $F = ma$. Instead, freely falling particles move along geodesics, which are the straightest possible paths in curved spacetime. Mathematically, geodesic motion is described by the geodesic

equation,

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0,$$

where τ is the proper time and $\Gamma_{\alpha\beta}^\mu$ are the Christoffel symbols constructed from the metric. These symbols encode how spacetime curvature alters particle trajectories. What appears as gravitational acceleration in Newtonian physics emerges naturally from spacetime geometry in General Relativity.

Mathematically, spacetime geometry is encoded in the metric tensor $g_{\mu\nu}$, which defines the spacetime interval between nearby events:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

Here, the coordinates $x^\mu = (ct, x, y, z)$ label events in spacetime, and the metric determines how distances and time intervals are measured. The metric in General Relativity can vary from point to point to reflect the influence of matter and energy on spacetime geometry.

The relationship between spacetime curvature and matter is governed by the Einstein field equations,

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

The left-hand side, represented by the Einstein tensor $G_{\mu\nu}$, describes the curvature of spacetime and is constructed from derivatives of the metric. The right-hand side, given by the stress-energy tensor $T_{\mu\nu}$, describes the distribution of energy, momentum, and pressure. This equation expresses the central idea of General Relativity: energy and momentum determine spacetime geometry.

To understand the content of the Einstein field equations more deeply, it is important to clarify the physical and mathematical meaning of both sides of the equation. Rather than being an arbitrary relation, this equation is carefully constructed so that each side encodes a specific aspect of gravity.

The left-hand side of the equation, $G_{\mu\nu}$, is known as the Einstein tensor and represents the geometric or curvature content of spacetime. Its role is to quantify how spacetime is curved by the presence of matter and energy. The Einstein tensor is built from the spacetime metric $g_{\mu\nu}$ and its derivatives, ensuring that it depends only on geometry rather than on any particular matter content. Conceptually, it describes how geodesics converge or diverge, which corresponds physically to tidal gravitational effects.

The Einstein tensor is defined as

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R,$$

where $R_{\mu\nu}$ is the Ricci curvature tensor and R is the Ricci scalar. The Ricci tensor itself is obtained by contracting the Riemann curvature tensor,

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu},$$

which encodes the full curvature information of spacetime. Physically, the Ricci tensor measures how the volume of a small bundle of geodesics changes as it moves through spacetime. The Ricci scalar $R = g^{\mu\nu} R_{\mu\nu}$ provides a single scalar measure of curvature by further contracting the Ricci tensor.

The specific combination used to form the Einstein tensor is not arbitrary. The subtraction of $\frac{1}{2}g_{\mu\nu}R$ ensures that the Einstein tensor satisfies the conservation condition

$$\nabla^\mu G_{\mu\nu} = 0,$$

which follows from the Bianchi identities of differential geometry. This property is essential because it guarantees consistency with local conservation laws for energy and momentum. In other words, spacetime curvature is constructed in a way that naturally respects conservation principles.

Physically, spacetime curvature manifests itself through tidal effects rather than through local forces. An observer in free fall experiences no gravitational force locally, but nearby free-falling particles may accelerate relative to one another. This relative acceleration is described by the geodesic deviation equation and is directly governed by spacetime curvature. In this sense, curvature is not detected by a single particle, but by comparing the motion of neighboring particles. This distinction is crucial in the early universe, where curvature arises from global energy distributions rather than from localized masses.

The right-hand side of the Einstein field equations, $T_{\mu\nu}$, is the stress-energy tensor and represents the energy-momentum content of spacetime. Unlike the Einstein tensor, which describes geometry, the stress-energy tensor describes the physical sources of gravity. It encodes mass-energy density, momentum flow, pressure, and stresses. This reflects a key insight of General Relativity: gravity is sourced by all forms of energy and momentum, not just rest mass.

For a relativistic perfect fluid, which is a good approximation for matter and radiation in cosmology, the stress-energy tensor takes the form

$$T^{\mu\nu} = (\rho + p/c^2)u^\mu u^\nu + pg^{\mu\nu},$$

where ρ is the energy density, p is the pressure, and u^μ is the four-velocity of the fluid. The first term represents energy and momentum carried by the fluid, while the

second term accounts for isotropic pressure. In the radiation-dominated early universe, pressure is comparable in magnitude to energy density, meaning it contributes significantly to the gravitational field.

The equality between the Einstein tensor and the stress-energy tensor therefore expresses the core principle of General Relativity: spacetime geometry responds directly to the energy and momentum of matter. The proportionality constant $\frac{8\pi G}{c^4}$ is chosen so that the theory reduces to Newtonian gravity in the weak-field, low-velocity limit. This limit, known as the Newtonian limit, corresponds to situations where gravitational fields are weak and velocities are much smaller than the speed of light. In this regime, the Einstein field equations reproduce Poisson's equation for the Newtonian gravitational potential, ensuring consistency with classical gravity.

This structure makes the Einstein field equations uniquely powerful. The left-hand side describes how spacetime is curved, the right-hand side describes what is causing that curvature, and the proportionality constant ensures agreement with known physics in appropriate limits. In the context of primordial black holes, this framework allows density fluctuations in the early universe to be treated self-consistently as sources of spacetime curvature, enabling the direct collapse of overdense regions into black holes without requiring stellar intermediaries.

In cosmological settings, matter and radiation are well described as a relativistic fluid. The stress-energy tensor for a perfect fluid takes the form

$$T^{\mu\nu} = (\rho + p/c^2)u^\mu u^\nu + pg^{\mu\nu},$$

where ρ is the energy density, p is the pressure, and u^μ is the four-velocity of the fluid. In the radiation-dominated early universe, pressure is comparable to energy density, which has important consequences for gravitational collapse. Unlike in Newtonian gravity, pressure contributes to the gravitational field, making collapse more difficult and raising the threshold required for PBH formation.

When the Einstein field equations are applied to a universe that is homogeneous and isotropic on large scales, they reduce to the Friedmann equations, which describe how the cosmic scale factor $a(t)$ evolves with time:

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho,$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right).$$

These equations show explicitly that both energy density and pressure influence cosmic expansion. During radiation domination, positive pressure slows expansion and

suppresses the growth of small density perturbations, which is why only sufficiently large overdensities can collapse to form primordial black holes.

A key relativistic concept relevant to PBH formation is the cosmic horizon, which defines the maximum distance over which causal interactions can occur. Gravitational collapse can only take place once a density perturbation enters the horizon, since regions larger than the horizon are not causally connected. This makes horizon re-entry a critical moment in determining whether an overdense region will disperse or collapse.

Black holes themselves are exact solutions of the Einstein field equations. The simplest such solution is the Schwarzschild metric, which describes the spacetime outside a spherically symmetric, non-rotating mass:

$$ds^2 = - \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2.$$

This solution predicts the existence of an event horizon at the Schwarzschild radius $r_s = 2GM/c^2$. If a mass is compressed to this radius, spacetime curvature becomes so extreme that nothing can escape, including light. This applies universally, regardless of whether the black hole forms from stellar collapse or from an early-universe density fluctuation.

It is important to emphasize that primordial black hole formation cannot be accurately described using Newtonian gravitational collapse alone. In Newtonian theory, collapse criteria such as the Jeans instability neglect relativistic effects, assume weak gravity, and treat pressure as opposing gravity rather than contributing to it. In the early universe, however, energy densities are extreme, pressure is relativistic, and the expansion of spacetime itself plays a dynamical role. General Relativity is therefore required to consistently describe the competition between gravitational attraction, pressure support, and cosmic expansion at horizon scales.

For primordial black holes, collapse is characterized by the density contrast

$$\delta = \frac{\delta\rho}{\rho}.$$

During radiation domination, simulations and analytical studies indicate that collapse requires δ to exceed a critical threshold of order 0.3–0.5. This relatively high threshold arises because radiation pressure acts rapidly to smooth out smaller overdensities. As a result, PBHs are rare and their abundance is extremely sensitive to the amplitude of primordial fluctuations.

The mass of a primordial black hole is set primarily by the horizon mass at the time of collapse,

$$M_H \sim \frac{c^3 t}{G},$$

with the PBH mass given by $M_{\text{PBH}} = \gamma M_H$, where γ is an efficiency factor of order 0.1. This relationship is central to our project, as it links early-universe dynamics to the mass scale of PBHs that could later serve as seeds for the supermassive black holes observed in JWST Little Red Dots.

In summary, General Relativity provides the framework needed to connect early-universe physics, horizon-scale dynamics, and black hole formation. By treating gravity as spacetime curvature rather than a force, GR explains why PBHs can form directly from primordial density fluctuations and why their initial masses are tightly linked to the expansion history of the universe. This makes GR an essential foundation for evaluating PBHs as viable seeds for the early supermassive black holes inferred in LRDs.